

# Assignment 3: 2D UAV Attitude, Translation, and Cascade Control

## Objective

Understand how a 2D UAV moves by tilting, how attitude stabilization works, and why cascade control requires a fast inner loop and a slower outer loop. This assignment connects Lecture 6 with the MATLAB code.

## Files to use

Use the modular version:

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>main_quadrotor2D_lab.m</b> | % run this file                 |
| <b>uav2d_params.m</b>         | % change parameters and gains   |
| <b>uav2d_controller.m</b>     | % inspect the control laws      |
| <b>uav2d_gain_cases.m</b>     | % define gain comparisons       |
| <b>uav2d_dynamics.m</b>       | % inspect the model             |
| <b>rk4_step.m</b>             | % inspect only; do not modify   |
| <b>animate_uav2d.m</b>        | % animation only; do not modify |
| <b>plot_uav2d_summary.m</b>   | % plotting only; do not modify  |

## Part 1 — Attitude control only

### Setup

In **main\_quadrotor2D\_lab.m**, set:

|                        |
|------------------------|
| scenario = 'attitude'; |
| compareGains = false;  |
| doAnimation = true;    |
| doPlots = true;        |

The UAV position is frozen. Only attitude dynamics are active:

$$I\ddot{\theta} = \tau$$

### Modify

In **uav2d\_params.m**, change:

|          |
|----------|
| p.att.kp |
|----------|

```
p.att.kd
```

In **main\_quadrotor2D\_lab.m**, change:

```
p.x0 = [0; 1; 0; 0; -25*pi/180; 0];  
p.ref.theta = 15*pi/180;
```

## Experiments

Run at least these cases:

- Nominal attitude control
  - Use the original gains.
  - Observe convergence of  $\theta$  to  $\theta_d$
- No damping
  - Set: **p.att.kd = 0;**
  - Observe oscillations.
- Weak attitude control
  - Reduce:
    - **p.att.kp**
    - **p.att.kd**
  - Observe slow response.
- Aggressive attitude control
  - Increase: **p.att.kp**
  - Observe faster response, overshoot, or actuator saturation.

## Report

1. Which gain gives the smoothest attitude response?
2. What happens when derivative damping is removed?
3. Does high proportional gain always improve the response?
4. Include one plot or screenshot from the animation.

## Part 2 — Translation control only

### Setup

In **main\_quadrotor2D\_lab.m**, set:

```
scenario = 'translation';  
compareGains = false;
```

Attitude is ignored. The UAV is treated as a point mass controlled by a virtual force.

## Modify

In **uav2d\_params.m**, change:

```
p.pos.kp  
p.pos.kd
```

In **main\_quadrotor2D\_lab.m**, change:

```
p.ref.pos = [3; 1];  
p.x0 = [0; 1; 0; 0; 0; 0];
```

## Experiments

Run at least these cases:

- Nominal position control
  - Use the original position gains.
- No velocity damping
  - Set: **p.pos.kd = [0; 0];**
  - Observe oscillations in position.
- High horizontal proportional gain
  - Increase: **p.pos.kp = [2.0; 1.0];**
  - Observe faster lateral motion and possible overshoot.
- Your best stable tuning
  - Choose gains that reach the target smoothly.

## Report

1. What is the role of p.pos.kp?
2. What is the role of p.pos.kd?
3. Why is this case not yet a realistic UAV model?
4. Include one plot of  $x(t)$  and  $z(t)$ .

## Part 3 — Full cascade control

### Setup

In **main\_quadrotor2D\_lab.m**, set:

```
scenario = 'cascade';
```

```
compareGains = true;
```

Now the outer loop generates a desired force, and the inner loop tracks the desired attitude.

## Inspect

In **uav2d\_controller.m**, find:

```
F = p.m*([0; p.g] + a_cmd);

T = norm(F);
theta_d = atan2(F(1), F(2));
theta_d = sat(theta_d, -p.thetaMax, p.thetaMax);

tau = p.att.kp*(theta_d - theta) - p.att.kd*q;
```

This is the key cascade connection:

$$F_d = \begin{bmatrix} F_x \\ F_z \end{bmatrix}, \quad T = \|F_d\|, \quad \theta_d = \text{atan2}(F_x, F_z)$$

## Modify

In **uav2d\_gain\_cases.m**, modify the three cases:

```
P(1).att.kp
P(1).att.kd
P(1).pos.kp
P(1).pos.kd

P(2).att.kp
P(2).att.kd

P(3).pos.kp
P(3).pos.kd
```

## Experiments

Run and compare:

- **Nominal cascade**

- Original gains.
- **Slow inner loop**
  - Reduce attitude gains:
    - **P(2).att.kp** = 0.08;
    - **P(2).att.kd** = 0.05;
  - Observe that position control becomes worse.
- **Aggressive outer loop**
  - Increase position gain: **P(3).pos.kp** = [2.00; 1.00];
  - Observe whether the attitude loop can keep up.
- **Your improved cascade**
  - Tune both loops.
  - Goal: stable motion, small overshoot, no strong oscillation.

## Report

1. Which case is most stable?
2. Which case oscillates most?
3. What happens when the outer loop is too aggressive?
4. What happens when the inner loop is too slow?
5. Compare  $\theta_d(t)$  and  $\theta(t)$ . Is the UAV able to track the commanded attitude?